

TOGI AK, AK, USA

WMO: 703606

Lat: 59.067N

Lon: 160.367W

Elev: 20

StdP: 14.69

Time Zone: -9.00 (NAK)

Period: 94-19

WBAN: 25629

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-17.5	-12.5	-27.3	1.2	-14.1	-25.6	1.3	-11.1	36.1	36.7	32.4	32.1	7.6	10	0.836

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.0	68.0	58.6	64.2	56.2	62.6	55.3	59.9	65.1	58.1	62.8	56.5	60.9	11.8	320

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
57.1	69.7	61.8	55.3	65.1	59.8	54.3	62.8	58.9	26.2	65.3	25.0	62.7	24.0	60.6	68.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
Min		Max		Min	Max		Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
31.9	27.0	24.0	DB	-23.5	74.7	5.8	3.9	-27.7	77.5	-31.1	79.8	-34.3	81.9	-38.5	84.7	
			WB	-23.4	61.4	4.6	3.0	-26.7	63.5	-29.4	65.3	-32.0	66.9	-35.3	69.1	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	34.6	17.8	20.0	19.5	31.4	41.9	50.2	54.0	53.1	47.3	36.0	24.2	18.4
	DBStd	17.05	14.99	14.37	12.66	7.92	6.14	4.84	4.04	3.83	5.00	8.32	11.70	14.99
	HDD50	5971	997	841	946	559	258	55	9	13	104	435	773	981
	HDD65	11112	1462	1261	1411	1009	716	445	342	367	531	899	1223	1446
	CDD50	336	0	0	0	0	7	60	133	111	24	1	0	0
	CDD65	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH74	7	0	0	0	0	1	4	1	1	0	0	0	0
	CDH80	1	0	0	0	0	0	1	0	0	0	0	0	0
Wind	WSAvg	11.0	11.3	12.4	11.6	11.5	10.6	9.6	9.9	10.3	10.8	11.1	11.2	12.1
Precipitation	PrecAvg	32.7	2.6	2.4	2.0	1.7	1.9	2.0	2.8	3.2	4.1	3.7	3.5	2.9
	PrecMax	48.0	6.0	6.4	5.6	4.0	3.6	3.5	4.5	5.1	7.8	7.2	6.9	5.0
	PrecMin	25.0	0.7	0.3	0.4	0.4	0.5	0.3	1.2	0.8	0.5	0.6	0.5	0.6
	PrecStd	5.3	1.1	1.3	1.3	0.8	0.8	0.7	0.9	1.1	1.6	1.5	1.7	1.3
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	43.0	39.4	43.6	52.8	67.9	72.6	72.1	70.0	62.9	53.7	49.2	43.0
		MCWB	37.9	36.5	37.8	43.1	52.8	57.8	61.4	59.2	53.2	47.9	45.4	38.4
	2%	DB	38.9	37.3	39.0	46.2	60.7	66.2	68.1	65.5	58.9	51.5	45.3	39.3
		MCWB	35.9	34.8	34.9	40.1	48.6	56.4	60.5	58.1	52.8	48.0	42.4	36.3
	5%	DB	36.8	36.4	36.6	43.4	54.8	63.0	64.2	63.0	56.7	48.5	42.8	37.2
		MCWB	34.5	34.3	33.6	38.1	47.1	54.7	57.1	56.3	51.7	46.0	40.1	35.0
	10%	DB	35.8	35.7	35.6	40.6	51.9	59.1	62.5	60.8	54.7	46.5	39.1	36.2
		MCWB	34.1	33.9	33.3	36.9	45.6	53.0	56.2	55.5	50.8	44.0	37.3	34.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	39.1	37.9	39.3	44.7	53.1	60.5	64.6	61.9	56.5	50.4	45.9	39.6
		MCDB	42.3	39.9	43.7	51.6	65.6	68.5	69.0	67.1	59.3	52.0	49.4	42.4
	2%	WB	36.5	35.8	35.6	41.0	50.0	57.8	60.8	59.3	54.3	48.3	43.1	37.2
		MCDB	38.4	36.8	38.1	46.0	57.7	64.3	66.3	63.4	56.9	50.3	45.3	38.9
	5%	WB	35.0	34.5	34.1	39.0	47.7	55.5	58.4	57.6	52.7	46.4	40.2	35.5
		MCDB	36.6	35.8	36.3	42.9	54.4	61.9	63.4	61.3	55.4	48.0	42.9	37.2
	10%	WB	33.5	33.3	32.8	37.2	46.1	53.2	56.7	56.2	51.3	45.2	37.7	34.2
		MCDB	35.2	34.9	34.8	40.2	50.6	58.2	61.3	59.8	54.2	46.7	39.2	35.9
Mean Daily Temperature Range	5% DB	MDBR	11.0	11.4	13.2	11.1	12.7	12.4	11.0	11.1	11.6	11.1	11.3	11.6
		MCDBR	7.0	8.1	10.2	14.5	21.9	20.8	18.1	16.6	14.2	9.3	8.2	7.4
		MCWBR	5.3	6.6	7.9	10.0	13.0	11.9	11.0	10.5	9.5	7.1	6.7	6.4
	5% WB	MCDBR	6.7	7.6	8.5	13.5	19.3	18.7	16.1	13.9	10.7	7.3	7.9	6.5
		MCWBR	5.5	6.6	6.8	9.5	11.8	11.3	10.6	9.4	8.0	6.4	6.7	5.9
Clear-Sky Solar Irradiance	taub	0.258	0.332	0.322	0.389	0.417	0.378	0.372	0.373	0.335	0.311	0.285	0.260	
	taud	2.112	2.031	2.115	2.006	2.106	2.354	2.415	2.411	2.488	2.378	2.101	1.982	
	Ebn at Noon	172	204	254	254	256	272	271	261	255	222	153	116	
	Edh at Noon	15	28	34	46	45	36	33	31	24	20	15	12	
All-Sky Solar Radiation	RadAvg	130	335	875	1339	1590	1637	1393	1094	773	412	160	72	
	RadStd	24	60	126	161	196	134	148	136	103	47	23	10	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46)	Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (5 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page